
CMP6200

Individual Undergraduate Project

Project Interim Report

**A Game-Based Approach to Motor Imagery
Adaptation and BCI Calibration**

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1.0 INTRODUCTION AND CONTEXT

1.1 Background and Significance

The way humans interact with computers has undergone a significant evolution over time. From early keyboards and mice to modern touchscreen devices and voice assistants, each innovation has expanded how people can engage with digital systems. But what if humans were able to communicate with technology simply by thought? This is what next-generation technology is achieving: Brain-Computer Interfaces (BCIs), systems that allow people to control devices with their thoughts.^(Birbaumer, 2006) By reading neural signals directly from the human brain and translating them into digital commands for computers to understand, BCIs not only enhance the experience for typical users but also enable people with physical limitations to access technology more easily.

Among various types of BCI applications, electroencephalography-based motor imagery (EEG-MI) is one of the most widely studied areas.^(Lotte et al., 2018) This concept is based on a key principle of neuroscience: whenever a person imagines a movement—such as moving a hand or foot—without performing it physically, their brain still produces characteristic signals with patterns that are remarkably similar to the actual physical execution.^(Neuper and Pfurtscheller, 2000) Building on this principle, EEG-based MI has been applied in many real-world scenarios, including motor rehabilitation, assistive control for individuals with disabilities, and control of games.^(Pfurtscheller and Lopes da Silva, 2001)

These promising applications, however, still depend heavily on how accurately MI signals can be detected. This accuracy is influenced by many factors, with the calibration process playing a crucial role.^(Lotte et al., 2018) Calibration—the supervised training phase where the system learns individual user neural patterns—remains a critical bottleneck limiting practical deployment, which is the main focus of this project.

1.2 Current State and Limitations

In controlled laboratory environments, where the calibration process is managed with precision, EEG-MI systems demonstrate a high level of performance in classifying motor signals.^(Lotte et al., 2018) After an initial training phase, these classifiers often reach accuracies above 70% in distinguishing between imagined movements such as left hand, right hand, foot, or a rest state, a level that is commonly taken as the threshold for effective BCI control.^(Kübler et al., 2001)

However, achieving this level of performance for each individual user requires a significant calibration effort. In practice, the system must learn a user's specific neural patterns through 15–30 minutes of supervised training, typically collecting 40–80 labelled trials per movement class (around 120–240 trials in total).^(Giles et al., 2022) Moreover, these EEG patterns vary significantly between sessions even for the same user due to non-

stationarities. (Sugiyama et al., 2008)

The repetitive and cognitively demanding nature of EEG-MI makes it challenging to implement outside controlled laboratory settings. (Xu et al., 2024) It is time-consuming, requires dedicated human supervision, and depends on specialised lab facilities (often geographically distant). It also induces cognitive fatigue, degrades signal quality over time, and limits practicality for multi-session applications like rehabilitation. (Xu et al., 2024) Consequently, despite decades of research, this technology remains largely confined to laboratories rather than achieving widespread real-world adoption. (Wolpaw et al., 2002)

1.3 Problem Statement

EEG-MI calibration per session requires 15–30 minutes of repetitive trials, which limits real-world scalability despite over 70% success in the experimental environment. (Giles et al.; Kueper et al., 2022; 2020) This limitation arises from changes in brain signals between sessions and across different individuals, (Sugiyama et al., 2008) the need for human supervision, (Giles et al., 2022) and reliance on specific infrastructure. (Lotte et al., 2018)

2.0 Review of Existing Knowledge

Comprehensively outline and expand on the primary concepts and established models presented in studies relevant to your problem statement. What is the existing knowledge on this subject? What are the main findings and conclusions derived from these studies? This should not merely be a descriptive summary; rather, a critical analysis of the information to construct a coherent narrative regarding the current state of knowledge.

Note: The review of concepts could be broken down into further sub-sections depending on the themes relevant to your topic that you identify during the review.

2.1 Critical Analysis and Gap Identification

During this phase, you carry out a critical reflection of the sources you have examined, identifying their limitations, contradictions, or disagreements within the literature. Establish which questions remain unanswered. This critical scrutiny should help you pinpoint a specific "gap"; whether it is an unresolved issue, an underexplored area, a new perspective that has not been considered, or a practical solution that is missing from existing studies.

2.2 Project Justification

Motor imagery-based BCIs are among the most commonly studied paradigms due to their non-invasive nature and potential for rehabilitation applications.

3.0 Project Aims, Objectives, and Scope

This section clearly defines what your project intends to achieve (aims), the specific steps you will take to reach those goals (objectives), and the boundaries within which your project will operate (scope).

3.1 Project Aims

The primary aim of this project is to develop a game-based motor imagery BCI system that can adapt to individual users, thereby enhancing calibration efficiency and overall user experience.

3.2 Project Objectives

To achieve the stated aim, the project will pursue the following objectives:

3.3 Project Scope

References

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